In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

In [3]:
```
df=pd.read_csv("student.csv")
```

In [4]:
```
df
```

Out[4]:

	student_id	weekly_self_study_hours	attendance_percentage	class_participation
0	1	18.5	95.6	3.8
1	2	14.0	80.0	2.5
2	3	19.5	86.3	5.3
3	4	25.7	70.2	7.0
4	5	13.4	81.9	6.9
...	...	...	...	...
999995	999996	18.0	95.5	4.8
999996	999997	15.7	82.7	6.7
999997	999998	14.2	85.1	5.8
999998	999999	25.3	90.0	5.3
999999	1000000	18.3	84.6	4.1

1000000 rows × 6 columns

In [5]:
```
df.head()
```

Out[5]:

	student_id	weekly_self_study_hours	attendance_percentage	class_participation	total_
0	1	18.5	95.6	3.8	
1	2	14.0	80.0	2.5	
2	3	19.5	86.3	5.3	
3	4	25.7	70.2	7.0	
4	5	13.4	81.9	6.9	

In [6]:
```
df.tail()
```

Out[6]:

	student_id	weekly_self_study_hours	attendance_percentage	class_participation
999995	999996	18.0	95.5	4.8
999996	999997	15.7	82.7	6.7
999997	999998	14.2	85.1	5.8
999998	999999	25.3	90.0	5.3
999999	1000000	18.3	84.6	4.1

In [7]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000000 entries, 0 to 999999
Data columns (total 6 columns):
 #   Column                   Non-Null Count    Dtype
---  ------                   --------------    -----
 0   student_id               1000000 non-null  int64
 1   weekly_self_study_hours  1000000 non-null  float64
 2   attendance_percentage    1000000 non-null  float64
 3   class_participation      1000000 non-null  float64
 4   total_score              1000000 non-null  float64
 5   grade                    1000000 non-null  object
dtypes: float64(4), int64(1), object(1)
memory usage: 45.8+ MB
```

In [8]:
```
df.describe()
```

Out[8]:

	student_id	weekly_self_study_hours	attendance_percentage	class_participati
count	1000000.000000	1000000.000000	1000000.000000	1000000.0000
mean	500000.500000	15.029127	84.711046	5.9852
std	288675.278933	6.899431	9.424143	1.9564
min	1.000000	0.000000	50.000000	0.0000
25%	250000.750000	10.300000	78.300000	4.7000
50%	500000.500000	15.000000	85.000000	6.0000
75%	750000.250000	19.700000	91.800000	7.3000
max	1000000.000000	40.000000	100.000000	10.0000

In [9]:
```
df.isnull()
```

Out[9]:

	student_id	weekly_self_study_hours	attendance_percentage	class_participation
0	False	False	False	False
1	False	False	False	False
2	False	False	False	False
3	False	False	False	False
4	False	False	False	False
...	...	...	...	...
999995	False	False	False	False
999996	False	False	False	False
999997	False	False	False	False
999998	False	False	False	False
999999	False	False	False	False

1000000 rows × 6 columns

In [10]:
```
df.isnull().sum()
```

Out[10]:
```
student_id                 0
weekly_self_study_hours    0
attendance_percentage      0
class_participation        0
total_score                0
grade                      0
dtype: int64
```

In [11]:
```
df.notnull()
```

Out[11]:

	student_id	weekly_self_study_hours	attendance_percentage	class_participation
0	True	True	True	True
1	True	True	True	True
2	True	True	True	True
3	True	True	True	True
4	True	True	True	True
...	...	...	...	...
999995	True	True	True	True
999996	True	True	True	True
999997	True	True	True	True
999998	True	True	True	True
999999	True	True	True	True

1000000 rows × 6 columns

In [12]:
```
df.ndim #no. of dimensions
```

Out[12]: 2

In [14]:
```
df.fillna
```

Out[14]:
```
<bound method NDFrame.fillna of         student_id  weekly_self_study_hours  at
tendance_percentage  \
0                1                     18.5                   95.6
1                2                     14.0                   80.0
2                3                     19.5                   86.3
3                4                     25.7                   70.2
4                5                     13.4                   81.9
...            ...                      ...                    ...
999995      999996                     18.0                   95.5
999996      999997                     15.7                   82.7
999997      999998                     14.2                   85.1
999998      999999                     25.3                   90.0
999999     1000000                     18.3                   84.6

        class_participation  total_score grade
0                       3.8         97.9     A
1                       2.5         83.9     B
2                       5.3        100.0     A
3                       7.0        100.0     A
4                       6.9         92.0     A
...                     ...          ...   ...
999995                  4.8        100.0     A
999996                  6.7         88.3     A
999997                  5.8         87.9     A
999998                  5.3        100.0     A
999999                  4.1        100.0     A

[1000000 rows x 6 columns]>
```

In [15]:
```
df.dropna
```

Out[15]:
```
<bound method DataFrame.dropna of         student_id  weekly_self_study_hours
attendance_percentage  \
0                1                     18.5                   95.6
1                2                     14.0                   80.0
2                3                     19.5                   86.3
3                4                     25.7                   70.2
4                5                     13.4                   81.9
...            ...                      ...                    ...
999995      999996                     18.0                   95.5
999996      999997                     15.7                   82.7
999997      999998                     14.2                   85.1
999998      999999                     25.3                   90.0
999999     1000000                     18.3                   84.6

        class_participation  total_score grade
0                       3.8         97.9     A
1                       2.5         83.9     B
2                       5.3        100.0     A
3                       7.0        100.0     A
4                       6.9         92.0     A
...                     ...          ...   ...
999995                  4.8        100.0     A
999996                  6.7         88.3     A
999997                  5.8         87.9     A
999998                  5.3        100.0     A
999999                  4.1        100.0     A

[1000000 rows x 6 columns]>
```

In [16]:
```
df.dtypes
```

Out[16]:
```
student_id                   int64
weekly_self_study_hours    float64
attendance_percentage      float64
class_participation        float64
total_score                float64
grade                       object
dtype: object
```

In [17]:
```
df['total_score'].min
```

Out[17]:
```
<bound method Series.min of 0          97.9
1           83.9
2          100.0
3          100.0
4           92.0
           ...
999995     100.0
999996      88.3
999997      87.9
999998     100.0
999999     100.0
Name: total_score, Length: 1000000, dtype: float64>
```

In [18]:
```
df['total_score'].median
```

Out[18]:
```
<bound method Series.median of 0          97.9
1           83.9
2          100.0
3          100.0
4           92.0
           ...
999995     100.0
999996      88.3
999997      87.9
999998     100.0
999999     100.0
Name: total_score, Length: 1000000, dtype: float64>
```

In [19]:
```
df['total_score'].mode
```

Out[19]:
```
<bound method Series.mode of 0          97.9
1           83.9
2          100.0
3          100.0
4           92.0
           ...
999995     100.0
999996      88.3
999997      87.9
999998     100.0
999999     100.0
Name: total_score, Length: 1000000, dtype: float64>
```

In [20]:
```
df['total_score'].std
```

Out[20]:
```
<bound method Series.std of 0          97.9
1          83.9
2         100.0
3         100.0
4          92.0
          ...
999995    100.0
999996     88.3
999997     87.9
999998    100.0
999999    100.0
Name: total_score, Length: 1000000, dtype: float64>
```

In [25]:
```
df.replace(["total_score"],["score"])
```

Out[25]:

	student_id	weekly_self_study_hours	attendance_percentage	class_participation
0	1	18.5	95.6	3.8
1	2	14.0	80.0	2.5
2	3	19.5	86.3	5.3
3	4	25.7	70.2	7.0
4	5	13.4	81.9	6.9
...	...	...	...	...
999995	999996	18.0	95.5	4.8
999996	999997	15.7	82.7	6.7
999997	999998	14.2	85.1	5.8
999998	999999	25.3	90.0	5.3
999999	1000000	18.3	84.6	4.1

1000000 rows × 6 columns

In [26]:
```
pd.get_dummies(df)
```

Out[26]:

	student_id	weekly_self_study_hours	attendance_percentage	class_participation
0	1	18.5	95.6	3.8
1	2	14.0	80.0	2.5
2	3	19.5	86.3	5.3
3	4	25.7	70.2	7.0
4	5	13.4	81.9	6.9
...	...	...	...	...
999995	999996	18.0	95.5	4.8
999996	999997	15.7	82.7	6.7
999997	999998	14.2	85.1	5.8
999998	999999	25.3	90.0	5.3
999999	1000000	18.3	84.6	4.1

1000000 rows × 10 columns

In [27]:
```
df.bfill()
```

Out[27]:

	student_id	weekly_self_study_hours	attendance_percentage	class_participation
0	1	18.5	95.6	3.8
1	2	14.0	80.0	2.5
2	3	19.5	86.3	5.3
3	4	25.7	70.2	7.0
4	5	13.4	81.9	6.9
...	...	...	...	...
999995	999996	18.0	95.5	4.8
999996	999997	15.7	82.7	6.7
999997	999998	14.2	85.1	5.8
999998	999999	25.3	90.0	5.3
999999	1000000	18.3	84.6	4.1

1000000 rows × 6 columns

In [28]:
```
df.ffill()
```

Out[28]:

	student_id	weekly_self_study_hours	attendance_percentage	class_participation
0	1	18.5	95.6	3.8
1	2	14.0	80.0	2.5
2	3	19.5	86.3	5.3
3	4	25.7	70.2	7.0
4	5	13.4	81.9	6.9
...	...	...	...	...
999995	999996	18.0	95.5	4.8
999996	999997	15.7	82.7	6.7
999997	999998	14.2	85.1	5.8
999998	999999	25.3	90.0	5.3
999999	1000000	18.3	84.6	4.1

1000000 rows × 6 columns

In [31]:
```
df.groupby('student_id')['total_score'].sum()
```

Out[31]:
```
student_id
1           97.9
2           83.9
3          100.0
4          100.0
5           92.0
           ...
999996     100.0
999997      88.3
999998      87.9
999999     100.0
1000000    100.0
Name: total_score, Length: 1000000, dtype: float64
```

In [ ]: